

MCQ Arena

PKI Acronyms: Full Forms and Definitions

Full forms, definitions, MCQ traps, memory lines, and certificate-security distinctions for fast PKI revision.

140	17
acronyms and key terms	revision sections

How to use this PDF

Start with the Most Important 20, then revise sections by function: certificates, hashing, algorithms, revocation, TLS, authentication, directories, hardware, standards, India PKI, and blockchain terms.

Core PKI and Certificate Terms

Acronym	Full form	Definition
PKI	Public Key Infrastructure	A framework of certificates, public/private keys, CAs, policies and procedures used to establish digital trust.
CA	Certificate Authority / Certifying Authority	A trusted entity that verifies identities and issues digitally signed certificates. In Indian law, the term Certifying Authority is commonly used.
RA	Registration Authority	A PKI entity that verifies the identity of certificate applicants before the CA issues certificates.
VA	Validation Authority	A service that helps determine whether a certificate is valid or revoked.
CSR	Certificate Signing Request	A signed request containing a public key and identity details, submitted to a CA for certificate issuance.
DSC	Digital Signature Certificate	An Indian PKI certificate used to associate a verified subscriber identity with a public key for digital signing.
DN	Distinguished Name	A structured X.509 name used to identify the subject or issuer of a certificate.
CN	Common Name	A DN attribute containing a person's name, server name or primary identity. Modern TLS hostname verification uses SAN rather than CN.
SAN	Subject Alternative Name	An X.509 extension containing certificate identities such as DNS names, IP addresses and email addresses.
CDP	CRL Distribution Point	A certificate extension identifying where clients can retrieve the issuer's CRL.
AIA	Authority Information Access	An extension that commonly provides the issuer-certificate or OCSP-responder location.
SKI	Subject Key Identifier	An identifier for the public key contained in a certificate.
AKI	Authority Key Identifier	An identifier that helps locate the issuer key that signed a certificate.
KU	Key Usage	An X.509 extension specifying permitted cryptographic operations, such as digital signing or certificate signing.
EKU	Extended Key Usage	An extension specifying application purposes such as server authentication, client authentication or code signing.
OID	Object Identifier	A globally unique number used to identify algorithms, certificate policies and X.509 extensions.
CPS	Certificate Practice Statement	A document explaining how a CA practically implements its certificate policies.
CP	Certificate Policy	A formal set of rules defining how a certificate may be issued, managed and used.

Hashing and Message Authentication

Acronym	Full form	Definition
SHA	Secure Hash Algorithm	A family of one-way cryptographic hash functions used to produce fixed-size message digests.
SHA-1	Secure Hash Algorithm 1	A 160-bit hash algorithm now considered insecure because practical collisions exist.
SHA-2	Secure Hash Algorithm 2	A secure hash family containing SHA-224, SHA-256, SHA-384 and SHA-512.
SHA-256	Secure Hash Algorithm, 256-bit	A SHA-2 algorithm producing a 256-bit digest, widely used in certificates and digital signatures.
SHA-512	Secure Hash Algorithm, 512-bit	A SHA-2 algorithm producing a 512-bit digest.
SHA-3	Secure Hash Algorithm 3	A newer hash family based on the Keccak construction.
HMAC	Hash-based Message Authentication Code	Combines a hash function with a shared secret key to provide message integrity and authentication.
MD5	Message Digest Algorithm 5	A 128-bit hash algorithm considered cryptographically broken because of collision attacks.
MAC	Message Authentication Code	A keyed value used to verify message integrity and authenticity.

Acronym	Full form	Definition
KDF	Key Derivation Function	A function that derives one or more cryptographic keys from a password, secret or shared value.
PBKDF2	Password-Based Key Derivation Function 2	Derives cryptographic keys from passwords using a salt and repeated computation.

MCQ distinction

Hash = Integrity only
HMAC = Integrity + Authentication
Encryption = Confidentiality

Cryptographic Algorithms

Acronym	Full form	Definition
AES	Advanced Encryption Standard	A fast symmetric block cipher used for bulk-data encryption.
DES	Data Encryption Standard	An obsolete symmetric block cipher using an effective 56-bit key.
3DES	Triple Data Encryption Standard	Applies DES three times; stronger than DES but now considered outdated.
RSA	Rivest-Shamir-Adleman	An asymmetric algorithm used for signatures and limited encryption or key transport; unsuitable for bulk encryption.
ECC	Elliptic Curve Cryptography	Public-key cryptography providing strong security with smaller keys than RSA.
ECDSA	Elliptic Curve Digital Signature Algorithm	An ECC-based algorithm used to create and verify digital signatures.
DH	Diffie-Hellman	A key-agreement method that allows two parties to establish a shared secret.
ECDH	Elliptic Curve Diffie-Hellman	An elliptic-curve version of Diffie-Hellman key agreement.
DHE	Ephemeral Diffie-Hellman	Diffie-Hellman using temporary keys to provide forward secrecy.
ECDHE	Elliptic Curve Diffie-Hellman Ephemeral	Temporary elliptic-curve key agreement used to provide forward secrecy.
RC4	Rivest Cipher 4	An obsolete symmetric stream cipher formerly used in TLS and wireless protocols.
RC5	Rivest Cipher 5	A configurable symmetric block cipher.
CSPRNG	Cryptographically Secure Pseudorandom Number Generator	Generates unpredictable random values suitable for cryptographic keys, salts and nonces.
RNG	Random Number Generator	A system that produces random or pseudorandom values.
PFS	Perfect Forward Secrecy	Protects previous sessions even if a server's long-term private key is later compromised.

Digital Signature Terms

Acronym	Full form	Definition
DSS	Digital Signature Standard	A U.S. standard specifying approved digital-signature algorithms.
DSA	Digital Signature Algorithm	A public-key algorithm designed specifically for digital signatures.
TSA	Time Stamping Authority	A trusted entity that issues digitally signed timestamp tokens proving that data existed at a particular time.
TSP	Time-Stamp Protocol	A protocol used to request and receive trusted timestamp tokens.
RFC 3161	Request for Comments 3161	The Internet specification defining a commonly used timestamp protocol.

Digital-signature key rule

Sign = Sender's private key
Verify = Sender's public key

Certificate Revocation

Acronym	Full form	Definition
CRL	Certificate Revocation List	A CA-signed list containing the serial numbers of certificates revoked before expiration.
Delta CRL	Delta Certificate Revocation List	A smaller CRL containing only status changes made since a particular base CRL.
OCSP	Online Certificate Status Protocol	Allows a client to query the status of an individual certificate and receive a signed response such as good, revoked or unknown.
OCSP Stapling	Online Certificate Status Protocol Stapling	A mechanism in which the server obtains and sends a signed OCSP response during the TLS handshake.

MCQ comparison

CRL = Download a complete revocation list
OCSP = Query the status of one certificate

Certificate Validation Levels

Acronym	Full form	Definition
DV	Domain Validation	Confirms control of the domain name but does not deeply verify the legal organization.
OV	Organization Validation	Verifies domain control and the identity of the organization.
EV	Extended Validation	Uses stricter organizational and legal-identity verification procedures.

Order of verification

DV < OV < EV

This represents increasing identity-verification depth, not stronger encryption.

Certificate Formats and Standards

Acronym	Full form	Definition
X.509	Not an acronym	The standard defining public-key certificates, certificate fields, extensions, CRLs and certification paths.
ASN.1	Abstract Syntax Notation One	A formal notation used to describe structures such as X.509 certificates.
DER	Distinguished Encoding Rules	A binary encoding of ASN.1 data used for certificates and keys.
PEM	Privacy-Enhanced Mail	A text representation containing Base64-encoded cryptographic data between header and footer lines.
CRT	Certificate	A common file extension for an X.509 certificate; it may contain PEM or DER data.
CER	Certificate	Another certificate file extension, commonly used in Windows environments.
PFX	Personal Information Exchange	A file extension commonly used for PKCS#12 containers containing a certificate and private key.
JKS	Java KeyStore	A Java-specific repository format for keys and certificates.

Acronym	Full form	Definition
CMS	Cryptographic Message Syntax	A standard format for signed and encrypted messages, developed from PKCS#7.

PKCS Standards

Acronym	Full form	Definition
PKCS	Public-Key Cryptography Standards	A family of standards defining cryptographic formats and interfaces.
PKCS#1	RSA Cryptography Standard	Defines RSA encryption, signatures and padding schemes such as OAEP and PSS.
PKCS#5	Password-Based Cryptography Standard	Defines password-based encryption and PBKDF2.
PKCS#7	Cryptographic Message Syntax predecessor	Defines signed/encrypted messages and certificate bundles; normally does not store private keys.
PKCS#8	Private-Key Information Syntax	A general format for storing private keys from different algorithms.
PKCS#10	Certification Request Syntax	Defines the standard CSR format.
PKCS#11	Cryptographic Token Interface, also called Cryptoki	Defines an API for interacting with HSMs, smart cards and cryptographic tokens.
PKCS#12	Personal Information Exchange Syntax	A password-protected container that may hold a private key, certificate and certificate chain.
PKCS#15	Cryptographic Token Information Format	Defines how cryptographic objects are organized on smart cards and tokens.
OAEP	Optimal Asymmetric Encryption Padding	A secure randomized padding scheme used with RSA encryption.
PSS	Probabilistic Signature Scheme	A randomized padding scheme used for modern RSA digital signatures.

Highest-value PKCS memory line

PKCS#1 = RSA
PKCS#5 = Password/PBKDF2
PKCS#7 = Messages and certificate bundles
PKCS#8 = Private-key format
PKCS#10 = CSR
PKCS#11 = HSM interface
PKCS#12 = Private key + certificate
PKCS#15 = Smart-card data organization

TLS and Website Security

Acronym	Full form	Definition
SSL	Secure Sockets Layer	An obsolete secure-communication protocol replaced by TLS.
TLS	Transport Layer Security	A protocol providing confidentiality, integrity and authentication for network communication.
HTTPS	Hypertext Transfer Protocol Secure	HTTP communication protected inside TLS.
HTTP	Hypertext Transfer Protocol	An application protocol used to exchange web content; it has no built-in encryption.
HSTS	HTTP Strict Transport Security	A web security policy that instructs browsers to access a domain only through HTTPS.
SNI	Server Name Indication	A TLS extension allowing the client to specify the requested hostname during the handshake.
AEAD	Authenticated Encryption with Associated Data	Encryption that provides confidentiality and integrity together.

Acronym	Full form	Definition
GCM	Galois/Counter Mode	An authenticated-encryption mode commonly used with AES.
CBC	Cipher Block Chaining	A block-cipher mode used in older TLS configurations.
mTLS	Mutual Transport Layer Security	TLS in which both the client and server authenticate using certificates.
ALPN	Application-Layer Protocol Negotiation	A TLS extension used to negotiate protocols such as HTTP/2.

Email Security

Acronym	Full form	Definition
PGP	Pretty Good Privacy	A system providing encryption and digital signatures, traditionally using a Web of Trust.
GPG	GNU Privacy Guard	A free implementation of the OpenPGP standard.
S/MIME	Secure/Multipurpose Internet Mail Extensions	A CA-based standard for digitally signing and encrypting email using X.509 certificates.
MIME	Multipurpose Internet Mail Extensions	Extends email to support HTML, different character sets and file attachments.
SMTP	Simple Mail Transfer Protocol	The protocol used to send and relay email.
SMTPS	Simple Mail Transfer Protocol Secure	SMTP started directly over an implicit TLS connection.
STARTTLS	Start Transport Layer Security	A command that upgrades an existing plaintext protocol connection to TLS.

Authentication and Access Control

Acronym	Full form	Definition
SFA	Single-Factor Authentication	Authentication using only one factor, such as a password.
MFA	Multi-Factor Authentication	Authentication using at least two different factor categories.
2FA	Two-Factor Authentication	MFA using exactly two different factor types.
SSO	Single Sign-On	Allows a user to authenticate once and access multiple connected applications.
AAA	Authentication, Authorization and Accounting	A framework for verifying identity, granting access and recording activity.
OTP	One-Time Password	A temporary code valid for one login or short period.
PIN	Personal Identification Number	A numeric secret used to unlock a device, smart card or authentication token.
FIDO	Fast Identity Online	Standards for phishing-resistant authentication using device-held public/private key pairs.
OIDC	OpenID Connect	An authentication and identity layer built on OAuth 2.0.
OAuth	Open Authorization	A delegated-authorization framework that grants limited access without sharing the user's password.
SAML	Security Assertion Markup Language	An XML-based enterprise federation and single-sign-on standard.
ZTA	Zero Trust Architecture	A security architecture based on explicit verification, least privilege and assuming breach.
RBAC	Role-Based Access Control	Grants permissions according to user roles.
ABAC	Attribute-Based Access Control	Makes access decisions using user, device, resource and environmental attributes.

Common MCQ traps

OIDC = Authentication
OAuth = Authorization
SSO = One login, multiple applications
MFA = Two or more different factor types

Authentication Protocols and Directories

Acronym	Full form	Definition
KDC	Key Distribution Center	The trusted Kerberos component that issues authentication tickets and session keys.
TGT	Ticket-Granting Ticket	A Kerberos ticket used to request service tickets without resending the password.
RADIUS	Remote Authentication Dial-In User Service	A centralized AAA protocol widely used for VPN, Wi-Fi and network access.
LDAP	Lightweight Directory Access Protocol	A protocol used to access and manage hierarchical directory information.
LDAPS	LDAP over SSL/TLS	LDAP communication protected by TLS, commonly associated with TCP port 636.
AD	Active Directory	Microsoft's enterprise directory service using technologies such as LDAP, Kerberos and DNS.
AD CS	Active Directory Certificate Services	Microsoft's enterprise PKI service for issuing and managing certificates.
DNS	Domain Name System	Translates domain names into IP addresses and supports service discovery.

Hardware and Key Protection

Acronym	Full form	Definition
HSM	Hardware Security Module	A tamper-resistant device used to generate, store and operate with cryptographic keys.
TPM	Trusted Platform Module	A hardware security component that protects keys and platform measurements.
USB	Universal Serial Bus	A hardware interface often used by cryptographic tokens and smart cards.
MDM	Mobile Device Management	A platform used to centrally manage mobile devices, certificates and security policies.
KMS	Key Management System / Key Management Service	A system for generating, storing, rotating and controlling cryptographic keys.

Standards and Organizations

Acronym	Full form	Definition
NIST	National Institute of Standards and Technology	A U.S. organization that publishes widely used security and cryptographic standards.
FIPS	Federal Information Processing Standards	U.S. government standards covering information-processing and cryptographic requirements.
FIPS 140-2	Federal Information Processing Standard Publication 140-2	A former standard defining four security levels for cryptographic modules.
FIPS 140-3	Federal Information Processing Standard Publication 140-3	The successor to FIPS 140-2 for validating cryptographic modules.
RFC	Request for Comments	A technical publication defining Internet protocols, formats and standards.
IETF	Internet Engineering Task Force	An organization that develops and publishes Internet standards through RFCs.
W3C	World Wide Web Consortium	An organization developing web standards such as WebAuthn.

Acronym	Full form	Definition
CAB Forum	CA/Browser Forum	An industry group that defines requirements for publicly trusted certificates.

India PKI and eSign

Acronym	Full form	Definition
CCA	Controller of Certifying Authorities	India's statutory authority that licenses and regulates Certifying Authorities.
UIDAI	Unique Identification Authority of India	The authority responsible for the Aadhaar identity system.
DSC	Digital Signature Certificate	A certificate used to verify a subscriber's digital-signature identity.
ESP	eSign Service Provider	An approved service provider facilitating regulated electronic-signature services.
ASP	Application Service Provider	An application platform that integrates an eSign service into a business workflow.
eSign	Electronic Signature	An electronic signing service; the term itself is not an acronym.
IT Act	Information Technology Act, 2000	India's primary statutory framework concerning electronic records, electronic signatures and cyber-related matters.

Blockchain Terms

Acronym	Full form	Definition
PoW	Proof of Work	A consensus method requiring computational effort to propose or validate blocks.
PoS	Proof of Stake	A consensus approach in which validators participate based on staked assets.
DLT	Distributed Ledger Technology	A replicated ledger maintained across multiple participating systems.
DApp	Decentralized Application	An application that uses blockchain or another decentralized network.

Most Important 20 for MCQs

Memorize these first:

PKI = Public Key Infrastructure
 CA = Certificate Authority
 CSR = Certificate Signing Request
 SHA = Secure Hash Algorithm
 HMAC = Hash-based Message Authentication Code
 CRL = Certificate Revocation List
 OCSP = Online Certificate Status Protocol
 DV = Domain Validation
 OV = Organization Validation
 EV = Extended Validation
 DSC = Digital Signature Certificate
 TSA = Time Stamping Authority
 PKCS = Public-Key Cryptography Standards
 FIPS = Federal Information Processing Standards
 MFA = Multi-Factor Authentication
 SSO = Single Sign-On
 KDC = Key Distribution Center
 RADIUS = Remote Authentication Dial-In User Service
 FIDO = Fast Identity Online
 ZTA = Zero Trust Architecture